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Enhancement of acidogenic fermentation for volatile fatty acid production from food waste: Effect of redox potential and inoculum

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HIGHLIGHTS

- Anaerobic and aerobic sludge were inoculated to food waste fermentation for VFA production under limited aeration condition.
- The maximal VFA yield was achieved at ORP in the range of −100 to −200 mV with anaerobic sludge as an inoculum.
- High protein degradation and VFA yield under limited aeration conditions were achieved.
- The low VFA production from food waste at pH 6.0 under anaerobic conditions was not caused by methanogenesis.

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ABSTRACT

The aim of this study was to explore the effects of redox potential (ORP) and inoculum on volatile fatty acids (VFAs) production from food waste by acidogenic fermentation. Four experimental conditions with two ORP levels were tested: limited aeration conditions with ORP level of −100 to −200 mV inoculating anaerobic sludge (LA + AnS) or aerobic sludge (LA + AeS), and anaerobic conditions with ORP level of −200 to −300 mV inoculating anaerobic sludge with 2-bromoethanosulfophate (AN + BES) and without BES (AN). The maximal VFA yield (0.79 g COD/g VS) was attained in LA + AnS reactor due to enhanced hydrolysis of substrates, especially proteins (degradation efficiency 78.3%). A higher frequency of phylum *Firmicutes* under limited aeration conditions (42.2–48.2%) was observed than that under anaerobic conditions (21.1%). The microbial community was more diverse in LA + AnS reactors than LA + AeS. We conclude that appropriate ORP level (from −100 to −200 mV) and inoculum play essential roles in VFA production.

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1. Introduction

At the first stage of a typical two-stage bioconversion process, hydrolysis and acidogenesis of organic matters in wastewater produce volatile fatty acids (VFAs) and other intermediates by mixed cultures (Li and Yu, 2011). VFAs (acetate, propionate, butyrate and valerate) have attracted a great deal of interests due to their wide potential applications, such as in biological nitrogen removal, bioplastics synthesis, and bioenergy production (Chen et al., 2013). Therefore, efficient conversions of organic matters into VFAs on acidogenic stage will improve bioenergy production. Generally, food waste (FW) with higher biodegradability is regarded as a desirable substrate for acidogenic fermentation.

Although VFAs offer promise in multiple industrial applications, the generation of VFAs by fermentation must overcome the limiting-rate step of hydrolysis and additionally the emergence of methanogenesis during anaerobic fermentation. By increasing the hydrolysis and acidification rates and meanwhile blocking the methane generation pathway, VFA production could be improved (Yuan et al., 2006). Strategies that have been used to enhance solubilization of organic matters into VFAs include physical/chemical pretreatments (Uellendahl and Kiaerahring, 2006), pH control (Yu and Fang, 2003), temperature control (Komemoto et al., 2009), and addition of enzymes (Cirne et al., 2008). In addition, the chemical 2-bromoethanosulfophate (BES) is often added to inhibit methanogenesis (Grootscholten et al., 2013).

In view of current global concerns over acidogenesis, most of the studies are operated under anaerobic conditions (Komemoto et al., 2009; Uellendahl and Kiaerahring, 2006; Yu and Fang, 2003). In fact, acid-forming bacteria (facultative or obligate anaer-

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obes) carry out hydrolysis, acidogenesis and acetogenesis, which have different nutritional needs and environmental conditions. Recently, several studies have shown that limited aeration in sludge and other organic substrates enhances methane production, by improving the hydrolytic-acidogenic phase (Xu et al., 2014b; Lim and Wang, 2013; Díaz et al., 2011). According to the present studies, low oxygen leads to a high yield and proportions of facultative acidogens (phylum *Firmicutes*, *Proteobacteria*, etc.), and high amounts of extracellular enzymes (Botheju and Bakker, 2011; Lim et al., 2014). However, due to different approaches to limited aeration for various organic wastes, the optimal limited aeration intensity remains unknown. Moreover, there is still a lack of studies concerning about acidogenesis under limited aeration conditions. In fact, oxygen content represents the redox information of fermentation system, and can be accurately indicated by the redox potential (ORP). ORP is an indicator of the net outcome of intracellular metabolism dominated by electron transfer and redox balance (Liu et al., 2013). According to Duangmanee (2009), the optimal ORP level for methanogenesis is below −230 mV. ORP values vary between −200 and −300 mV under fully anaerobic conditions (Henze, 2008). But based on Xu et al. (2014a) finding, the optimal ORP range for acidogenic fermentation of food waste in leach bed reactor is between −100 mV and −250 mV, which is achieved by aeration. However, the detailed information about the role of ORP in hydrolytic-acidogenic process has not been shown. Besides, the mechanism about how ORP level influences VFA production from FW has not been studied. In this study, we used ORP to characterize the oxygen-limited and fully anaerobic environments, with the idea that this would reveal how ORP level affected the acidogenic fermentation of FW.

In addition, the type of inoculum is a significant factor affecting the evolution of fermentative pathways (De Gioannis et al., 2013). In a previous study, we find higher VFA production can be obtained by inoculating anaerobic granular sludge than aerobic sludge for FW fermentation (Wang et al., 2014). However, the work do not reveal whether differences in inoculum under limited aeration conditions account for the higher VFA production.

Consequently, in order to investigate the role of ORP level in FW acidogenic reactors, two ORP levels of −100 to −200 mV (oxygen-limited) and −200 to −300 mV (anaerobic) were set to reveal how ORP level affect the hydrolytic-acidogenic phase of FW fermentation. Two inocula were used to estimate the impact of microbial composition on acidogenesis under limited aeration conditions. To determine the effect of BES on VFA accumulation, we also tested the VFA production with or without addition of BES under anaerobic conditions. We hypothesized that ORP level of −100 to −200 mV would benefit VFA production by altering the microbial community. Here we showed the microbial community analysis, which could be used to understand the relationship between ORP level, inoculum, hydrolysis, and acidogenesis in acidogenesis from FW.

2. Methods

2.1. Substrate and inocula

Simulated food waste, containing rice (36%), noodle (26%), cabbage (22%), tofu (14%), and pork (2%) was compounded based on the characteristics of FW in our previous study (Yin et al., 2014). The five components came from the same vendor at Cui yuan farmers' market (Hangzhou, China) and were crushed using a mangler. The simulated food waste was then stored frozen at −18 °C. Before use, the FW was treated by hydrothermal (HT) pretreatment, as described by the previous study (Yin et al., 2014). Two inocula were used: aerobic activated sludge from the secondary settling tank of the Qige Wastewater Treatment Plant and anaerobic granular sludge from an up-flow anaerobic sludge blanket (UASB) reactor of the Xihu Brewery (Hangzhou, China). Both of the two inocula would have to be reactivated with culture medium before inoculating. The main characteristics of the FW (before and after hydrothermal treatment) and two inocula were listed in Table 1.

2.2. Experimental set-up

The material in this study consisted of a mixture of 28 g pretreated simulated food waste and 7 g inoculum (dry weight) was in amber wide-mouth bottles (F/M ≈ 5.0 g VS/g VS). The working volume of a reactor was 500 mL and tap water was added to attain total solids (TS) of 70 g/L. Four operating conditions existed in this study (Table 2). The reactors were operated under limited aeration conditions by inoculating anaerobic sludge (LA + AnS) or aerobic sludge (LA + AeS), and under anaerobic conditions by inoculating anaerobic sludge with 2-bromoethanosulfophate (AN + BES) and without BES (AN). By means of limited aeration, ORP could be maintained in the range of −100 to −200 mV (Xu et al., 2014a; Lim et al., 2014). So air was added to the reactor at rate of 274 L air/kg TS/d (according to our previous preliminary test), where ORP were measured per 2 h to maintain its level between −100 and −200 mV. Nitrogen gas was flushed through the reactors to create anaerobic conditions, keeping ORP level in the ranged of −200 to −300 mV. BES, when added, was 10 mM final concentration in the reactor. During the fermentation, all of the reactors stirred mechanically at 120 rpm using a magnetic stirrer. The experimental temperature was controlled at 30 ± 2 °C and the pH at 6.0 by adding 4.5 M HCl or NaOH (based on our previous study (Wang et al., 2014)). Each experiment was performed in duplicate reactors for 21 days.

2.3. Analysis methods

2.3.1. Chemical analysis

Soluble chemical oxygen (SCOD), soluble sugar, soluble protein, ammonia nitrogen (NH_4^+-N), and VFAs contents were all determined every 2 days. Total solids (TS), volatile solids (VS), volatile suspended solids (VSS), total nitrogen (TN), total protein (T-protein), total sugar (T-sugar), lipids (T-lipids) and total chemical oxygen (TCOD) were measured before and after fermentation. The fermented broth was separated from the residue by centrifuging at 11,000×g for 5 min and filtered using the microfiltration membrane (0.45 μm).

ORP values were recorded using an ORP meter (Leici Corp., Shanghai, China). TS, VS, VSS, TN, NH_4^+-N, TCOD and SCOD were all analyzed using standard methods (APHA, 1998). T-protein was estimated from the TN concentration (T-protein = TN * 6.25) (Yuan et al., 2006). Lipids were extracted by Soxhlet extraction

Table 1
Characteristics of FW before and after hydrothermal treatment and two inocula.

Parameter	Original FW	HT-FW	An-sludge[a]	Ae-sludg[b]
TS (%)[d]	25.7	24.2	11.3	3.0
VS (%)[d]	98.2	99.5	78.0	84.4
T-protein (%)[d]	15.1	17.1	29.3	8.5
TCOD (%)[d]	1.1	1.4	1.3	1.0
T-sugar (%)[d]	0.6	0.6	0.1	0.1
T-lipids (%)[d]	0.03	0.03	–[c]	0.02
SCOD (g/L)	87.8	197.5	23.5	14.5
S-sugar (g/L)	35.9	147.8	0.1	2.7
S-protein (g/L)	1.5	11.9	–[c]	3.0

[a] An-sludge: anaerobic activated sludge.
[b] Ae-sludge: aerobic activated sludge.
[c] "–":undetected.
[d] %: the proportion of TS/VS/T-protein/TCOD/T-sugar/T-lipids in per g total dry sample.

Table 2
Operating conditions for acidogenic fermentation.

ORP (mV)	Oxygen conditions	Inoculum	Inhibitor	Reactor
−100 to −200	Limited aeration	Anaerobic sludge	/	LA + AnS
−100 to −200	Limited aeration	Aerobic sludge	/	LA + AeS
−200 to −300	Anaerobic	Anaerobic sludge	BES	AN + BES
−200 to −300	Anaerobic	Anaerobic sludge	/	AN

BES: 2-bromoethanosulfophate.

(Liu, 1994). Soluble protein was quantified by the Lowry–Folin method using bovine serum albumin as the standard (Lowry et al., 1951), and carbohydrate was determined by using the phenol sulfuric acid method with glucose as the standard (Herbert et al., 1971). Volatile fatty acids (VFAs, C2–C5), including acetate, propionate, *n*-butyrate, *iso*-butyrate, *n*-valerate, and *iso*-valerate, were determined using a GC7890-II gas chromatograph (Tianmei Co., Shanghai, China) equipped with a 3 m × 2 mm stainless steel packed column filled with GDX-103 as the stationary phase and a flame ionization detector. The temperatures of the column, injector, and detector were 180, 230, and 250 °C, respectively.

Protein, sugar and VFAs concentrations were expressed as the COD equivalent, calculated based on the complete oxidation of the individual matter to CO_2 and H_2O. The COD conversion factors of protein, sugar, acetate, propionate, butyrate and valerate were 1.50, 1.06, 1.07, 1.51, 1.82 and 2.04, respectively.

2.3.2. Enzyme activity measurement

The activities of key hydrolytic enzymes (protease and α-glucosidase) were determined according to the method developed by Goel et al. (1998). The specific enzyme activity was defined as unit of enzyme activity per gram of VS.

2.3.3. Bacterial community analysis

Microbial samples were acquired from all reactors at the outset and end of the reaction, as well as the timepoint with maximum VFA production, for bacterial community analysis. The genomic DNA was extracted using an kit (DR4011; Bioteke Corporation, Beijing, China) according to the manufacturer's instructions. The quality (A260/A280) and quantity (A260) of the extracted genomic DNA were determined with a Nanodrop 2000c spectrophotometer (Thermo Fisher Scientific, Waltham, MA). The extracted DNA was amplified using PCR reactions and the universal 16S rRNA primers F (CAAGCAGAAGACGGCATACGAGATGTGACTGGAGTTCAGACGTGTGCTCTTCCGATCT (barcode) ACTCCTACGGGAGGCAGCAG) and R (AATGATACGGCGACCACCGAGATCTACACTCTTTCCCTACACGACGCTCTTCCGATCT (barcode) GGACTACHVGGGTWTCTAAT). The bacterial community was determined by Illumina high-throughput sequencing. Sequences were aligned and grouped into operational taxonomic units (OTUs) with 97% similarity. Sequences were then phylogenetically assigned to taxonomic classifications and were allocated into phylum, class and genus levels. Hierarchical cluster analysis of all samples was performed using R version 3.1.3.

3. Results and discussion

3.1. VFA production from food waste

3.1.1. Effect of redox potential (ORP) levels

Two operating conditions (LA + AnS and AN) with different ORP levels for VFA production were tested (Table 2). During acidogenic fermentation, ORP levels were monitored (Fig. 1). As shown in Fig. 1, under LA + AnS conditions ORP levels were stabilized between −100 mV and −200 mV after limited oxygen was added. The ORP levels under AN conditions kept between −200 and −300 mV. To investigate whether ORP levels affect the VFA production by fermentation, VFA concentrations in LA + AnS and AN reactors were measured. As shown in Fig. 2a, on day 3, nearly two times higher yield of VFAs was observed in LA + AnS reactors than that in AN. And then VFA accumulation increased rapidly to about 20 g/L on day 9, which was still two times of that in AN reactors. The maximal VFA concentration was 29.4 g/L obtained on day 17. However, VFA accumulation increased at a slow rate in AN reactors before day 7. And then the highest VFA production (18.36 g/L) was achieved on day 17, indicating the negative effect of anaerobic conditions (ORP ranged from −200 to −300 mV) on acidogenesis. Jagadabhi et al. (2010) and Xu et al. (2014a) also find the positive effect of limited aeration on VFA production. Low oxygen leads to a high yield and proportions of facultative acidogens (phylum *Firmicutes*, *Proteobacteria*, etc.) and improves the hydrolytic-acidogenic phase (Botheju and Bakker, 2011; Xu et al., 2014b). Compared with other previous studies (Table 3), VFA yield obtained under LA + AnS conditions (ORP ranged from −100 to −200 mV) was 1.3 times higher than that in Xu's study.

Among the VFAs produced in LA + AnS and AN reactors, 80% of VFAs consisted of H-Ac and *n*-Bu, where *n*-Bu accounted for up to 60% (Fig. 2b). We attributed this pattern to the >40% total carbohydrate contents of FW in this study, which was consistent with our previous reports (Wang et al., 2014; Yin et al., 2014). In addition, another two VFAs (*iso*-Bu, H-Va) were apparently found in LA + AnS reactors but were less than 1 g/L in AN. This result showed ORP levels also would affect the VFA composition, and there were more kinds of acidogens under LA + AnS conditions.

3.1.2. Effect of inoculum

Combining the results of the ORP levels with VFA production, we found that an ORP between −100 and −200 mV for the whole process enhanced VFA yield from FW by acidogenic fermentation. To investigate whether inoculum affected the VFA production, anaerobic sludge and aerobic sludge as inocula were compared under limited aeration conditions. VFA concentrations in LA + AnS and LA + AeS reactors were measured. Under LA + AeS conditions, ORP values were above −100 mV before day 6 (Fig. 1), which could be caused by residual oxygen and the decay of aerobic microorganisms in the aerobic sludge. As shown in Fig. 2a, in LA + AeS reactors, VFA accumulation did not increase before day 7,

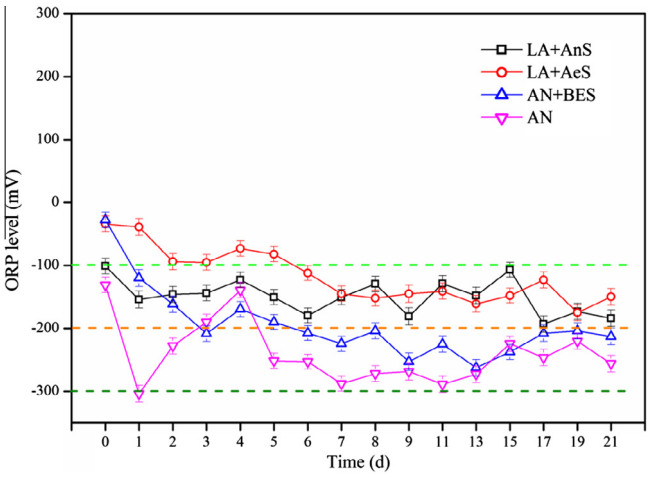


Fig. 1. ORP level of four conditions during acidogenic fermentation. According to the change tendency of ORP in all these reactors, the values were displayed per day during 0–9 d, but every two days after day 9.

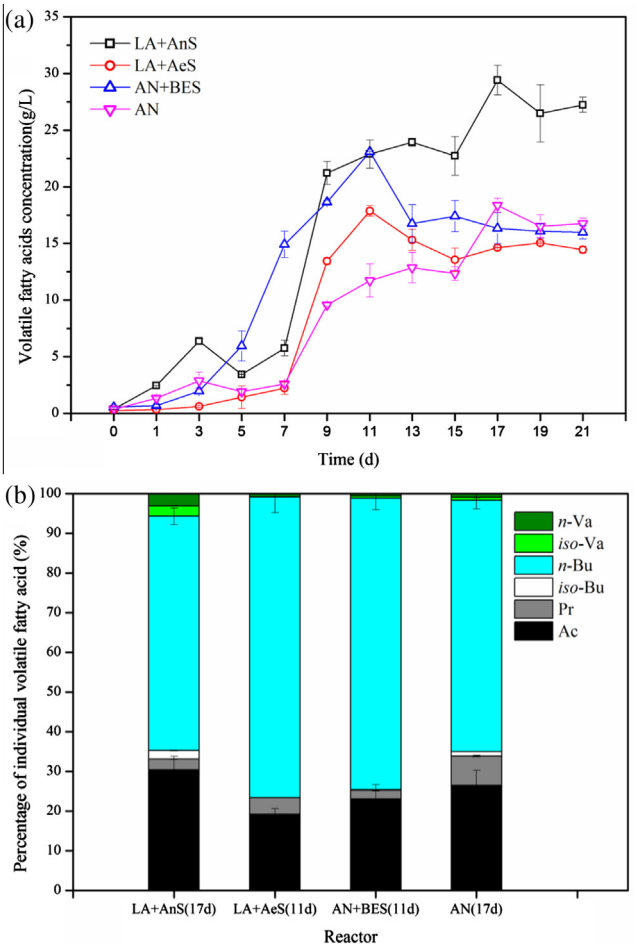


Fig. 2. (a) VFA production and (b) VFA compositions for four conditions during fermentation. The VFA composition analysis was based on the maximum VFA production of each reactor: LA + AnS (17d), LA + AeS (11d), AN (17d), AN + BES (11d).

which was attributed to less acidogens in aerobic sludge and higher ORP. After day 7, the VFA production in LA + AeS reactors increased fast, and ORP declined around −100 to −200 mV due to the accumulation of facultative anaerobes. On day 11, VFA concentration achieved the maximum 17.98 g/L, but was still less than that in LA + AnS (29.4 g/L). Therefore, under limited aeration conditions, AnS as an inoculum obviously showed superiority over AeS in acidogenesis. Compared to LA + AnS condition, among VFAs, *iso*-Bu and H-Va also accounted for little in LA + AeS reactors (Fig. 2b). This phenomenon was mostly related to the different microbial composition caused by the inocula.

3.1.3. Effect of BES addition

BES was added to determine whether VFA production could be enhanced by inhibiting the methanogenesis under anaerobic conditions, so two operating conditions (AN + BES and AN) for VFA production were compared. As shown in Fig. 1, the ORP levels under AN conditions kept between −200 and −300 mV. However, when BES was added, the ORP values were generally higher (between −100 and −200 mV) before day 6, and then dropped to below −200 mV. The higher ORP levels in the first 6 days could be attributed to that BES (as an extra electron acceptor) addition would increase the level of oxidation of reactor (Chiu and Lee, 2001). After day 6, some of BES were consumed to weaken this effect, making ORP dropped. Until recently, the knowledge about the relationship between ORP level and BES addition was still limited. For VFA production, reactor AN + BES accelerated acid yields after day 5, and maximal VFA concentration (23.09 g/L) was obtained on day 11 (Fig. 2a). The VFA production in AN + BES reactors was higher than that in AN, which correlated with its ORP level between −100 and −200 mV before day 6. ORP decreased and VFA accumulated slowed after day 6. The other possible reason for higher VFA production in AN + BES reactors was that methanegenesis was inhibited by BES to promote the VFA accumulation. However, there was little methane production (less than 5%) in 21-day fermentation (data not shown). In addition, BES addition had no effect on VFA composition (Fig. 2b).

From the above results, we concluded that less negative ORP levels (−100 to −200 mV) would play a positive role in VFA production. Under limited aeration conditions, inoculating anaerobic sludge would benefit higher VFA yield from more facultative acidogens. In addition, BES addition indeed improved VFA accumulation in acidogenic fermentation. Reasons for all these phenomena were analyzed as follows.

3.2. Hydrolysis of organic matters for VFA production from FW

3.2.1. Carbohydrates solubilization

VFA production is closely related to hydrolysis and degradation of substrates. Soluble sugar reflects a balance between rates of solubilization and degradation (Fig. 3a). On day 1, concentrations of soluble sugar decreased in LA + AnS, AN, AN + BES reactors, but increased in LA + AeS reactors. It inferred that solubilization dominated in LA + AeS. Differently, in other reactors, degradation of carbohydrates played the dominant role from the beginning of fermentation. After day 1, soluble sugar concentrations decreased rapidly in all reactors. However, the uptake rates of soluble sugar were different in each reactor as follows: FW_{AN} (6.3 g/L·d) < FW_{AN+BES} (6.4 g/L·d) < FW_{LA+AnS} (7.4 g/L·d) < FW_{LA+AeS} (8.8 g/L·d). The reason for higher degradation rates in LA + AnS and LA + AeS reactors was assumed to be the presence of higher amounts of extracellular enzymes under limited aeration conditions (Botheju and Bakker, 2011). From day 7, soluble carbohydrates were almost consumed completely in all reactors. Moreover, the conversion efficiencies of total sugar in all reactors were above 88.8% at the end of fermentation, indicating there was no obviously selective degradation for sugar with different ORP levels or inocula.

Table 3
Comparison of VFA production from different limited aeration intensities.

Material	Aeration intensity	ORP level (mV)	VFA production	Reference
Food waste	258 L air/kg TS/d	−100 to −250	0.39 g COD/g VS (17d)	Xu et al. (2014a)[a]
Grass silage	2.5 L air (1 L/min)	–	0.33 g COD/g VS (35d)	Jagadabhi et al. (2010)[b]
Food waste	274 L air/kg TS/d	−100 to −200	0.79 g COD/g VS (17d)	This study

[a] Xu, S.Y., Selvam, A., Wong, J.W., 2014a. Optimization of micro-aeration intensity in acidogenic reactor of a two-phase anaerobic digester treating food waste. Waste Mange. 34, 363–369.
[b] Jagadabhi, P., Kaparaju, P., Rintala, J., 2010. Effect of micro-aeration and leachate replacement on COD solubilization and VFA production during mono-digestion of grass-silage in one-stage leach-bed reactors. Bioresour. Technol. 101, 2818–2824.

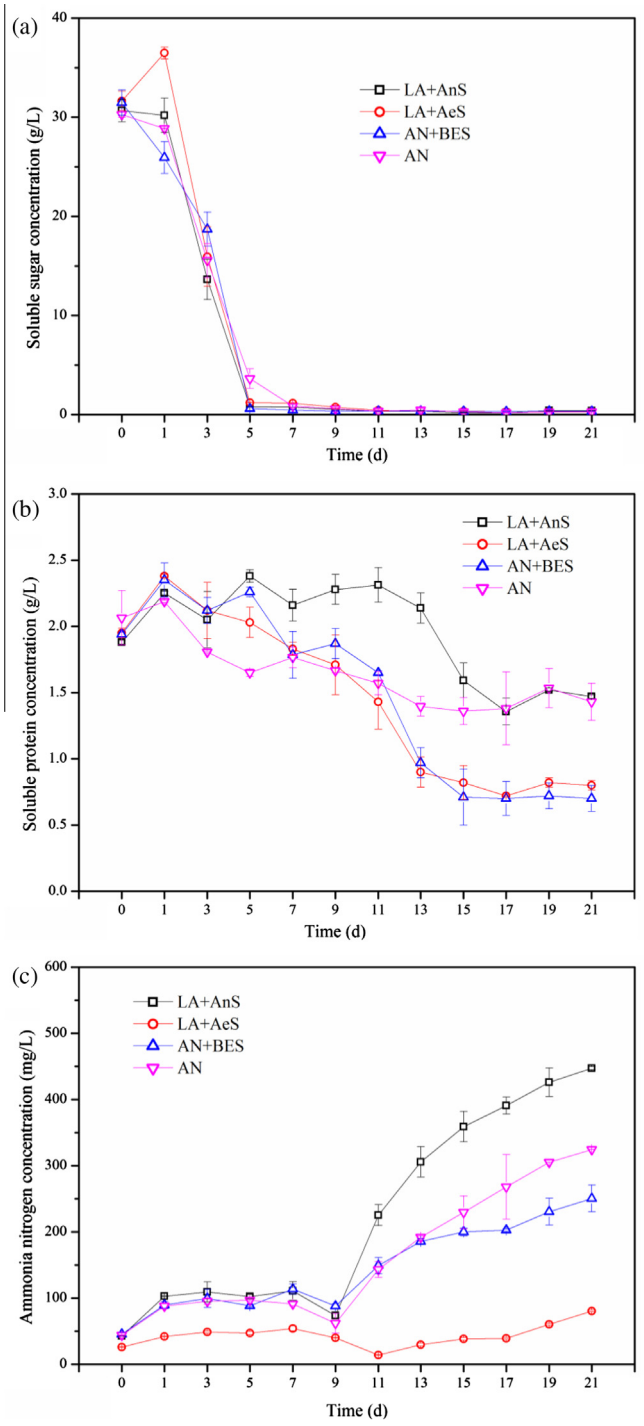


Fig. 3. (a) the solubilization of sugar, (b) protein, (c) ammonia nitrogen release for four conditions during acidogenesis.

Table 4
Conversion of components in FW before and after fermentation.

Sample	VS[a]	T-COD[a]	T-sugar[a]	T-protein[a]	T-lipids[a]	Biomass[a]
Original	33.3	49.2	17.8	6.81	0.95	5.46[b]
LA + AnS_{end}	12.4	18.9	1.89	1.48	0.18	5.46[b]
LA + AeS_{end}	13.3	18.5	1.76	3.95	0.36	5.46[b]
AN_{end}	14.0	18.0	1.70	3.88	0.58	5.46[b]
AN + BES_{end}	13.6	24.7	1.54	2.62	0.32	5.46[b]

[a] Unit: g (the content of VS/TCOD/T-sugar/T-protein/T-lipids/Biomass in the whole reactor).
[b] Due to the slow growth of fermentative microorganisms, here it was assumed that the content of biomass (the volatile solid content of inoculum) kept unchanged during the whole fermentation.

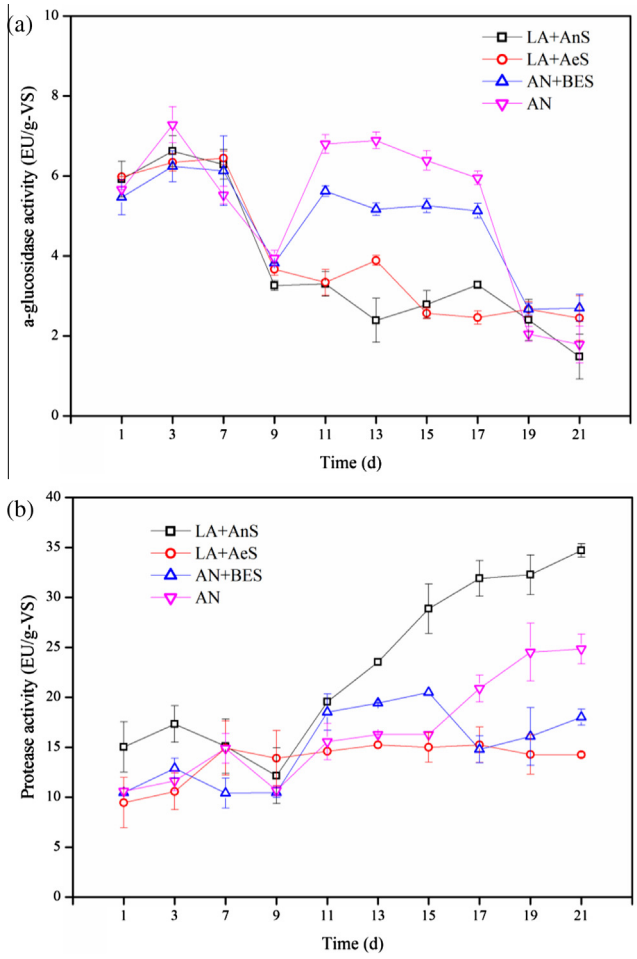


Fig. 4. (a) a-glucosidase, (b) protease activities for four conditions during acidogenic fermentation.

3.2.2. Proteins solubilization and ammonia nitrogen release

During the whole fermentation process, there were downward trends for soluble proteins under all conditions (Fig. 3b). LA + AnS reactors had the most soluble proteins. Soluble protein concentrations in AN reactors remained relatively stable during the fermentation and just decreased from 2000 to 1200 mg/L. These results demonstrated that less negative ORP levels (−100 to −200 mV) could promote proteins solubilization. It was likely that limited aeration environment could stimulate the effective co-existence of facultative bacteria and anaerobes (Sang et al., 2009) and low oxygen promoted fermentative bacteria to excrete enzymes (Lim et al., 2014). After day 15, the concentration levels of soluble proteins almost kept unchanged under all conditions, which indicated there was a balance between dissolution and degradation of protein. Interestingly, at the end of fermentation, some soluble proteins were still undegraded in all reactors.

The proteins in FW are first degraded to amino acids, then to ammonia, VFAs and other products. Consequently, ammonia (NH_4^+-N) is generally used to assess the level of protein degradation. For all conditions, NH_4^+-N concentrations remained low and stable before day 9 (Fig. 3c). After day 9, NH_4^+-N concentration increased approximately linearly with fermentation time except LA + AeS condition. The release rates of NH_4^+-N were: FW_{LA+AeS}

(3.38 mg L^{-1} d^{-1}) < FW_{AN+BES} (13.5 mg L^{-1} d^{-1}) < FW_{AN} (21.9 mg L^{-1} d^{-1}) < FW_{LA+AnS} (31.1 mg L^{-1} d^{-1}). Higher release rate of NH_4^+-N in LA + AnS reactors also showed that less negative ORP level (from −100 to −200 mV) enhanced the degradation of soluble protein. At the end of fermentation, the degradation efficiencies of proteins were 78.3%, 42.0%, 43.0% and 61.5% in LA + AnS, LA + AeS, AN and AN + BES reactors, respectively (Table 4). Specially, lowest protein transformation rate was observed under LA + AeS conditions due to less facultative bacteria and microbial diversity in aerobic sludge (see Section 3.4).

Based on the above results, we assured that less negative ORP levels (−100 to −200 mV) had a positive effect on substrates degradation, especially proteins. And inoculating anaerobic sludge would benefit substrates transformation. The higher organics degradation efficiency led to more VFA accumulation.

3.3. The activities of key enzymes

Carbohydrates firstly hydrolyze into monosacchride, meanwhile proteins hydrolyze into amino acids, and then they all turn into pyruvate. So we next measured activities of α-glucosidase and protease under four conditions during the acidogenic fermentation. As shown in Fig. 4a, there were the same trends of α-glucosidase activities for all reactors before day 9. With the decrease of carbohydrates in FW, α-glucosidase activities for all reactors decreased to about 4.0 U/g VS on day 9. It was likely that carbohydrates were hydrolyzed completely, reducing enzyme activities. But afterwards, α-glucosidase activities increased in AN and AN + BES reactors. On the contrary, they continued to decline in LA + AnS and LA + AeS reactors. Since soluble sugars were almost completely degraded quickly before day 9, the impact of α-glucosidase activity increased was unclear.

Apparently, during the whole fermentation, protease activities in LA + AnS reactors were higher than those in other reactors, which explained the phenomenon described above that soluble protein concentration was higher and degradation rate was faster under LA + AnS conditions. The improved hydrolysis could be attributed to the stimulation of fermentative bacteria to excrete hydrolytic enzymes under micro-aeration condition (Xu et al., 2014a). After day 15, the protease activities for all reactors were as follows: $FW_{LA+AnS} > FW_{AN} > FW_{AN+BES} > FW_{LA+AeS}$. Overall, protease activity in LA + AeS reactors was the lowest throughout the fermentation. It was supposed that hydrolysis of proteins was the rate-limiting step in acidogenic fermentation (Wang et al., 2016), and aerobic sludge was not suitable to be used as an inoculum for acidogenesis from FW.

3.4. The microbial structure analysis

We speculated that the differences in VFA productions and enzyme activities of these reactors were related to the structure of microbial community. 16S rRNA high-throughput sequencing was performed to determine the contribution of the microbial community to the reactor outcome. We identified 32 phyla in LA + AnS reactors, 28 in AN + BES, 25 in LA + AeS, and 22 in AN. The majority of sequences belonged to phyla *Firmicutes*, *Proteobacteria*, and *Bacteroidetes* for all reactors at the time of maximum VFA production (Fig. 5a). The sum of these three phyla accounted for 71.3% (LA + AnS), 83.4% (AN + BES), 84.3% (LA + AeS) and 93.5% (AN) of the total reads. This result showed the higher microbial diversity under LA + AnS conditions. Phylum *Firmicutes* was the highest under limited aeration conditions (42.2% for LA + AnS, 48.2% for LA + AeS) and the lowest in AN reactors (21.1%). However, compared to LA + AnS (11.0%) and LA + AeS (0.3%), phylum *Bacteroide-*

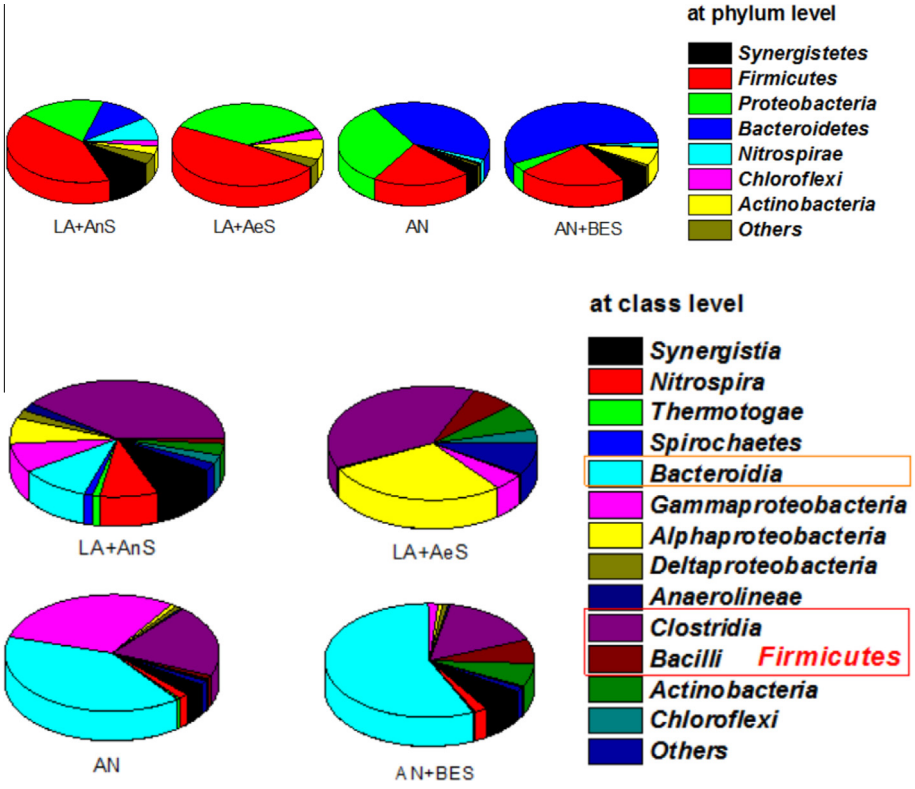


Fig. 5. Taxonomic classification of MiSeq sequencing from bacterial community of LA + AnS, LA + AeS, AN, AN + BES samples at phylum and class. Relative abundance was defined as the number of sequences affiliated with that taxon divided by the total number of sequences per sample. Phyla and classes making up less than 1% of the total composition in all reactors were classified as "others".

tes was more plentiful in AN (40.1%) and AN + BES (57.0%) reactors. Phylum *Firmicutes* is facultative bacteria, which is known to produce cellulases, lipases, proteases and other extracellular enzymes (Leven et al., 2007). *Bacteroidetes* have been reported that they can exist in obligate fermentation and acidification bioreactors (Lim et al., 2014). Interestingly, phylum *Chloroflexi* was absent under anaerobic conditions. Most of *Chloroflexi* are filamentous bacteria, and they have the ability to degrade macromolecular organics (Yu et al., 2016).

At a class level, most of the reads (119 classes identified in total) belonged to 13 classes (Fig. 5b), mainly with four classes (*Clostridia*, *Alphaproteobacteria*, *Gammaproteobacteria* and *Bacteroidia*). *Clostridia* (about 40%) within *Firmicutes* was the dominant class under limited aeration conditions, but under anaerobic conditions there were a great number of *Bacteroidia* within *Bacteroidetes* (more than 35%).

Genus level characterization showed that the initial samples of three systems (LA + AnS, AN, AN + BES) were clustered together, indicating the microbial community structure were similar in these reactors at the start of fermentation because of the same inoculum AnS (Fig. 6). During the acidogenesis, a remarkable increase in microbial diversity was observed in LA + AnS reactor. Larger amounts of *Anaerofilum* (30.8%), *Clostridium* (4.1%), *Pseudomonas* (3.9%), *Prevotella* (5.8%), *Oscillospira* (2.6%), *Ruminococcus* (2.3%), *Brevundimonas* (1.7%) were detected under this condition. *Clostridium, Oscillospira*, *Anaerofilum* and *Ruminococcus* within *Firmicutes* have been shown to dominate FW acidogenic reactors (Grootscholten et al., 2013; Mgrootscholten et al., 2014). The

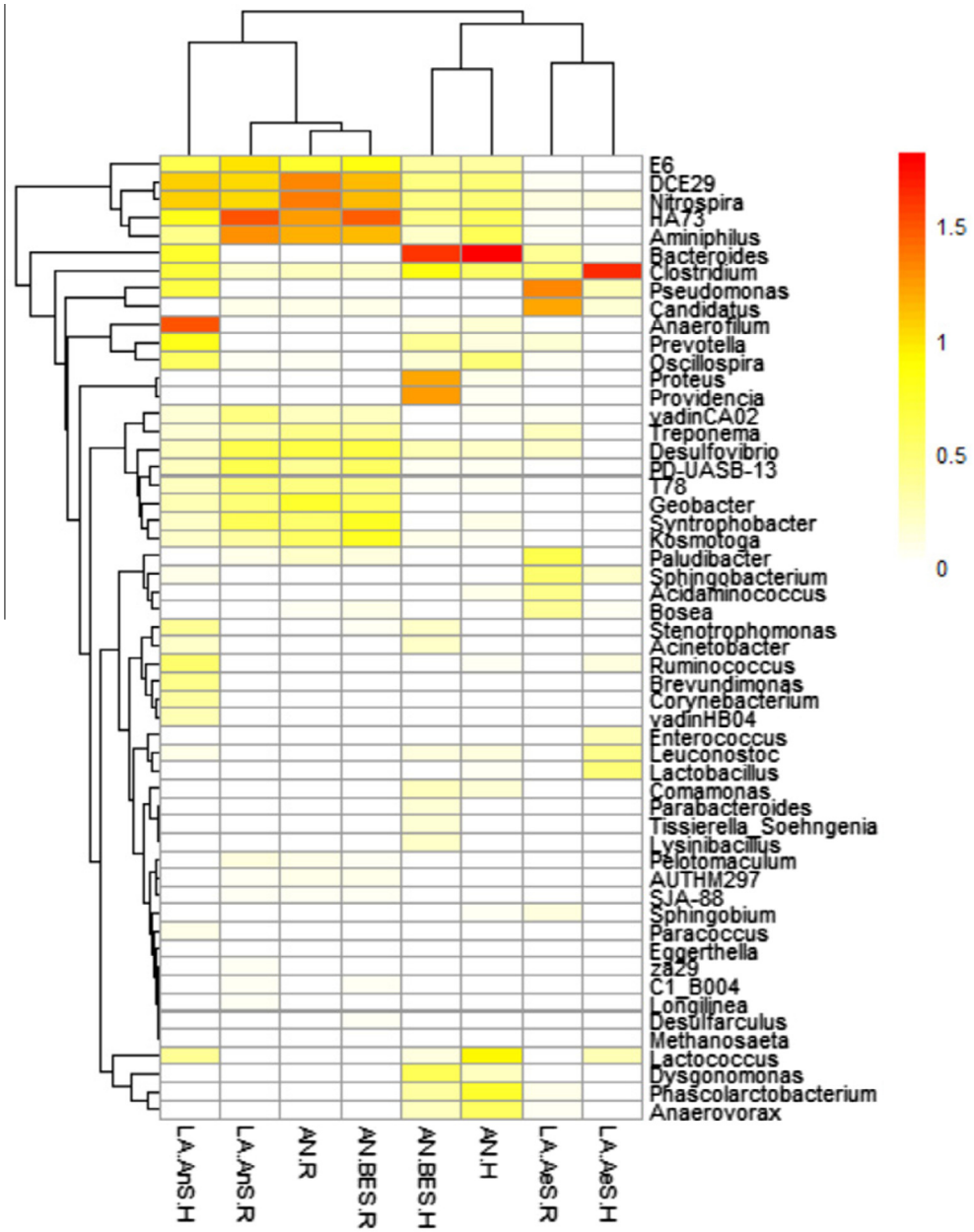


Fig. 6. Hierarchical cluster analysis of bacterial communities in LA + AnS, LA + AeS, AN and AN + BES samples. The y-axis is the 54 most abundance genus in reads under all conditions. Samples (the initial of fermentation and the maximum VFA yield) were clustered based on complete linkage method, the color intensity showed the relative abundance of each genus. Relative abundance was defined as the number of sequences affiliated with that genus divided by the total number of sequences per sample. LA. AnS.R (LA.AeS.R, AN.R, AN.BES.R)indicated the initial samples in LA + AnS (LA + AeS, AN, AN + BES) reactors; LA.AnS.H (LA.AeS.H, AN.H, AN.BES.H) indicated samples with maximum VFA yield in LA + AnS (LA + AeS, AN, AN + BES) reactors.

dominance of *Firmicutes* indicates the ability of acidogenic reactor to metabolize various substrates, such as proteins, carbohydrates, and lipids (Lim et al., 2014). *Pseudomonas*, *Prevotella and Brevundimonas* previously demonstrate high ability of acidogenesis. Therefore, more soluble substrates and VFA production in LA + AnS reactors were correlated with the predominance of *Firmicutes* and some other acidogens. Paradoxically, although *Clostridium* also accounted for higher proportion (45.0%) in LA + AeS reactors, the VFA yield was the least on account of the less microbial diversity and less *Clostridium* in absolute amount in aerobic sludge for FW acidogenesis.

By contrast, the abundance of *Bacteroides* apparently increased in AN (63.6%) and AN + BES (40.9%) reactors, a reflection of the anaerobic environment. Nevertheless, presumably because of the presence of BES, a clear distinction of microbial community diversity was present in these two anaerobic reactors. Some studies find that long-term exposure to BES not only eliminate Archaea but also alter the bacterial community structure by acting as a competing electron acceptor for some microorganisms (Chiu and Lee, 2001; Ye et al., 1999). Significantly, the abundance of *Methanogenus* was less than 1% under both AN and AN + BES conditions. There were some other dominant genera such as *Proteus* (15.0%), *Providencia* (17.2%), *Prevotella* (1.4%) in AN + BES reactors, which were hardly detected in AN. Higher VFA production in AN + BES mainly was attributed to the wide existence of *Proteus* and *Providencia* that could degrade organics to acids. These results implied that the low VFA production under anaerobic conditions was not caused by methanogenesis.

4. Conclusions

The optimal ORP level for VFA production was −100 to −200 mV, which could be obtained by limited aeration during acidogenic fermentation. At this ORP level, anaerobic sludge as an inoculum promoted the degradation of substrates, especially protein, and thus enhanced VFA yield. Anaerobic condition (ORP −200 to −300 mV) was not suitable for VFA production at pH 6.0. Higher VFA yield appeared to depend on the greater abundance of *Firmicutes* and more diverse microbial community under LA + AnS conditions. BES addition in acidogenesis would improve VFA yield by increasing abundance of some acidogens rather than its inhibition to methanogenesis.

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References

APHA, 1998. Standard Methods for the Examination of Water and Wastewater, 20th ed. American Public Health Association/American Water Works Association/ Water Environment Federation, Washington, DC, USA.
Botheju, D., Bakker, R., 2011. Oxygen effects in anaerobic digestion–a review. Open Waste Mange. J. 4, 1–19.
Chiu, P.C., Lee, M., 2001. 2-Bromoethanesulfonate affects bacteria in a trichloroethene-dechlorinating culture. Appl. Environ. Microbiol. 67, 2371–2374.
Chen, H., Meng, H.J., Nie, Z.C., Zhang, M.M., 2013. Polyhydroxyalkanoate production from fermented volatile fatty acids: Effect of pH and feeding regimes. Bioresour. Technol. 128, 533–538.
Cirne, D.G., Agbor, V.B., Bjoernsson, L., 2008. Enhanced solubilisation of the residual fraction of municipal solid waste. Water Sci. Technol. 57 (7), 995–1000.
De Gioannis, G., Muntoni, A., Polettini, A., Pomi, R., 2013. A review of dark fermentative hydrogen production from biodegradable municipal waste fractions. Waste Manage. (Oxford) 33 (6), 1345–1361.
Díaz, I., Donoso-Bravo, A., Fdz-Polanco, M., 2011. Effect of microaerobic conditions on the degradation kinetics of cellulose. Bioresour. Technol. 102, 10139–10142.
Duangmanee, T., 2009. Micro-Aeration for Hydrogen Sulfide Removal From Biogas. Iowa State University, Ph.D.
Goel, R., Mino, T., Satoh, H., Matsuo, T., 1998. Enzyme activities under anaerobic and aerobic conditions in activated sludge sequencing batch reactor. Water Res. 32 (7), 2081–2088.
Grootscholten, T.I.M., Steinbusch, K.J.J., Hamelers, H.V.M., Buisman, C.J.N., 2013. Chain elongation of acetate and ethanol in an upflow anaerobic filter for high rate MCFA production. Bioresour. Technol. 135, 440–445.
Herbert, D., Philipps, P.J., Strange, R.E., 1971. Carbohydrate analysis. Method Enzymol. 5B, 265–277.
Henze, M., 2008. Biological Wastewater Treatment. IWA Publishing.
Jagadabhi, P., Kaparaju, P., Rintala, J., 2010. Effect of micro-aeration and leachate replacement on COD solubilization and VFA production during mono-digestion of grass-silage in one-stage leach-bed reactors. Bioresour. Technol. 101, 2818–2824.
Komemoto, K., Lim, Y.G., Nagao, N., Onoue, Y., Niwa, C., Toda, T., 2009. Effect of temperature on VFAs and biogas production in anaerobic solubilization of food waste. Waste Mange. 29, 2950–2955.
Leven, L., Ieriksson, A., Schnurer, A., 2007. Effect of process temperature on bacterial and archaeal communities in two methanogenic bioreactors treating organic household waste. FEMS Microbiol. Ecol. 59 (3), 683–693.
Li, W.W., Yu, H.Q., 2011. From wastewater to bioenergy and biochemicals via two-stage bioconversion processes: A future paradigm. Biotechnol. Adv. 29, 972–982.
Lim, J.W., Wang, J.Y., 2013. Enhanced hydrolysis and methane yield by applying micro-aeration pretreatment to the anaerobic co-digestion of brown water and food waste. Waste Manage. 33, 813–819.
Lim, J.W., Chiam, J.A., Wang, J.Y., 2014. Microbial community structure reveals how microaeration improves fermentation during anaerobic co-digestion of brown water and food waste. Bioresour. Technol. 171, 132–138.
Liu, K.S., 1994. Preparation of fatty-acid methyl esters for gas-chromatographic analysis of lipids in biological materials. J. Am. Oil Chem. Soc. 71 (11), 1179–1187.
Liu, C.G., Xue, C., Li, Y.H., Bai, F.W., 2013. Redox potential control and application in microaerobic and anaerobic fermentations. Biotechnol. Adv. 31 (2), 257–265.
Lowry, O.H., Rosebrough, N.J., Farr, A.L., Randall, R.J., 1951. Protein measurement with the Folin phenol reagent. J. Biol. Chem. 193, 265–275.
Mgrootscholten, T., Bstrik, D., Jsteinbusch, K., Nbuisman, C., Mhamelers, H., 2014. Two-stage medium chain fatty acid (MCFA) production from municipal solid waste and ethanol. Appl. Energ., 223–229
Sang, N.N., Soda, S., Inoue, D., Sei, K., Ike, M., 2009. Effect of intermittent and continuous aeration on accelerative stabilization and microbial population dynamics in landfill bioreactors. J. Biosci. Bioeng. 108 (4), 336–343.
Uellendahl, H., Kiaerahring, B., 2006. Strategies for the anaerobic digestion of the organic fraction of municipal solid waste. Water Sci. Technol. 53 (8).
Wang, K., Yin, J., Shen, D.S., Li, N., 2014. Anaerobic digestion of food waste for volatile fatty acids (VFAs) production with different types of inoculum: effect of pH. Bioresour. Technol. 161, 395–401.
Wang, X., Li, Y.B., Liu, J.X., Ren, N.Q., Qu, J.H., 2016. Augmentation of protein-derived acetic acid production by heat-alkaline-induced changes in protein structure and conformation. Water Res. 88, 595–603.
Xu, S.Y., Selvam, A., Wong, J.W., 2014a. Optimization of micro-aeration intensity in acidogenic reactor of a two-phase anaerobic digester treating food waste. Waste Manage. 34, 363–369.
Xu, S., Ammaiyappan, S.O., Karthikeyan, O.P., Wong, J.W.C., 2014b. Responses of microbial community and acidogenic intermediates to different water regimes in a hybird solid anaerobic digestion system treating food waste. Bioresour. Technol. 168, 49–58.
Ye, D., Quensen, J.F., Tiedje, J.M., Boyd, S.A., 1999. 2-Bromoethanesulfonate, sulfate, molybdate, and ethanesulfonate inhibit anaerobic dechlorination of polychlorobiphenyls by pasteurized microorganisms. Appl. Environ. Microbiol. 65 (1), 327–329.
Yin, J., Wang, K., Yang, Y.Q., Shen, D.S., Wang, M.Z., Mo, H., 2014. Anaerobic digestion of food waste for volatile fatty acid production with different types of inoculum: Effect of pH. Bioresour. Technol. 171, 323–329.
Yu, H.Q., Fang, H.H.P., 2003. Acidogenesis of gelatin-rich wastewater in an upflow anaerobic reactor: influence of pH and temperature. Water Res. 37 (1), 55–66.
Yu, H., Fang, C.H., Liu, X.P., Yi, X.Y., Ren, Y., Wei, C.H., 2016. Enhanced anaerobic dechlorination of polychlorinated biophenyl in sediments by bioanode stimulation. Environ. Pollut. 211, 81–89.
Yuan, H.Y., Chen, Y.G., Zhang, H.Y., Jiang, S., Zhou, Q., Gu, G.W., 2006. Improved bioproduction of short-chain fatty acids (SCFAs) from excess sludge under alkaline conditions. Environ. Sci. Technol. 40, 2025–2029.